This unit is assessed by means of a written examination paper of 1 hour 35 minutes duration. The paper will consist of objective, short-answer and long-answer questions. Students may be required to apply their knowledge and understanding of physics to situations that they have not encountered before. The total number of marks available for this examination paper is 80. It contributes 40% to IA2 and 20% to the IAL in Physics.

It is recommended that students have access to a scientific calculator for this paper.

Students will be provided with the formulae sheet shown in *Appendix 5*. Any other physics formulae that are required will be stated in the question paper.

The quality of written communication will be assessed in the context of this unit through questions which are labelled with an asterisk (*). Students should take particular care with spelling, punctuation and grammar, as well as the clarity of expression, on these questions.

3.3 Further Mechanics

This topic covers momentum and circular motion.

This topic may be studied using applications that relate to, for example, a modern rail transport system.

Students will be assessed on their ability to:		Suggested experiments
73	use the expression $p = mv$	
74	investigate and apply the principle of conservation of linear momentum to problems in one dimension	Use of, for example, light gates and air track to investigate momentum
75	investigate and relate net force to rate of change of momentum in situations where mass is constant (Newton's second law of motion)	Use of, for example, light gates and air track to investigate change in momentum
76	derive and use the expression $E_k = p^2/2m$ for the kinetic energy of a non-relativistic particle	
77	analyse and interpret data to calculate the momentum of (non-relativistic) particles and apply the principle of conservation of linear momentum to problems in one and two dimensions	
78	explain and apply the principle of conservation of energy, and determine whether a collision is elastic or inelastic	
79	express angular displacement in radians and in degrees, and convert between those units	
80	explain the concept of angular velocity, and recognise and use the relationships $v = \omega r$ and $T = 2\pi/\omega$	
81	explain that a resultant force (centripetal force) is required to produce and maintain circular motion	
82	use the expression for centripetal force $F = ma = mv^2/r$ and hence derive and use the expressions for centripetal acceleration $a = v^2/r$ and $a = r\omega^2$	Investigate the effect of m, v and r of orbit on centripetal force